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Track C – Applied Optimization

1. Problem & Domain

I selected the Supply Chain Risk & Optimization scenario and implemented a prediction-driven optimization workflow for inventory allocation. The objective is to determine how limited inventory should be allocated across multiple branches while accounting for uncertain future demand, supplier risk, transportation costs, and warehouse-capacity constraints. Demand forecasting is used as an input to optimization because forecasts alone do not prescribe actions. A synthetic dataset representing 100 SKUs across 8 branches over 36 months was generated, including seasonality, inventory levels, supplier risk, transportation costs, warehouse capacities, lead times, and stockout events.

2. Approach & Algorithm Decisions

The solution consists of two stages. First, demand forecasting was performed using engineered features such as lag demand, rolling statistics, demand growth metrics, inventory ratios, and calendar features. Two models were evaluated: Random Forest and XGBoost. Random Forest served as a baseline, while XGBoost was selected for its ability to model nonlinear interactions. ARIMA and LSTM were considered but rejected due to limited support for operational features and increased complexity. XGBoost achieved the lowest RMSE and was selected as the optimizer input.

Second, inventory allocation was formulated as a linear optimization problem. The decision variable represents the quantity of each SKU allocated to each branch. The objective maximizes profit while accounting for transportation costs and supplier-risk penalties. Constraints include inventory availability, warehouse capacity, service-level targets, and demand satisfaction. PuLP with the CBC solver was selected because it provides a transparent and reproducible optimization framework.

3. Results & Error Analysis

Random Forest achieved RMSE = 6.24, MAE = 2.03, and MAPE = 2.47%. XGBoost achieved RMSE = 5.51, MAE = 2.76, and MAPE = 3.90%. Since RMSE is most relevant for downstream optimization, XGBoost was selected. The optimization model solved approximately 800 allocation decisions across 100 SKUs and 8 branches. Results were: Status = Optimal, Runtime less than one second, Profit = \$5.91M, and Constraint Satisfaction Rate = 100%. Sensitivity analysis showed profitability is more sensitive to demand changes than moderate warehouse-capacity changes. The primary source of error is forecasting uncertainty, which can propagate into allocation decisions.

4. Production Considerations & Limitations

In production, the forecasting and optimization pipeline would integrate with ERP and inventory-management systems. Key failure modes include forecast drift and optimization infeasibility caused by conflicting constraints. Mitigation strategies include automated retraining, forecast monitoring, soft constraints, and penalty-based relaxation. The current implementation uses deterministic forecasts, linear cost assumptions, and synthetic data. Future enhancements include probabilistic forecasting, stochastic optimization, experiment tracking, and model explainability.